

Scenario Summary Table — Problem 4

5 scenarios · 10 instances each · 50 total · all values from generator code

| Parameter                 | S1                                         | S2                    | S3                   | S4                   | S5                         |
|---------------------------|--------------------------------------------|-----------------------|----------------------|----------------------|----------------------------|
| General parameters        |                                            |                       |                      |                      |                            |
| Name                      | Baseline                                   | High low-level demand | Geographic imbalance | High order load      | Tight relocation budget    |
| Factor tested             | All neutral                                | F1 — demand level mix | F2 — flow imbalance  | F3 — order/car ratio | F1+F3 combined + 4th level |
| n_S — stations            | 14                                         | 16                    | 18                   | 20                   | 20                         |
| n_C — cars                | 45                                         | 54                    | 60                   | 72                   | 72                         |
| n_L — levels              | 3                                          | 3                     | 3                    | 3                    | 4                          |
| n_K — orders              | 145                                        | 145                   | 150                  | 150 ■                | 190                        |
| n_D — days                | 8                                          | 8                     | 9                    | 9                    | 10                         |
| B — budget (min)          | 6,000                                      | 8,400                 | 12,000 ■             | 15,000 ■             | 4,500                      |
| Order/car ratio           | 3.2x                                       | 2.7x                  | 2.5x                 | 2.1x ■               | 2.6x                       |
| Duration (h)              | 2-16                                       | 2-16                  | 2-14                 | 2-14                 | 3-18                       |
| Pick-up window            | Days 1-7                                   | Days 1-7              | Days 1-8             | Days 1-8             | Days 1-9                   |
| Moving time (min)         | 60-240                                     | 60-240                | 60-240               | 60-210               | 90-300                     |
| Seeds                     | 4901-4910                                  | 5001-5010             | 5101-5110            | 5201-5210            | 5301-5310                  |
| Fleet composition         |                                            |                       |                      |                      |                            |
| Level-1 cars              | 18                                         | 22                    | 24                   | 29                   | 26                         |
| Level-2 cars              | 18                                         | 22                    | 24                   | 29                   | 22                         |
| Level-3 cars              | 9                                          | 10                    | 12                   | 14                   | 15                         |
| Level-4 cars              | —                                          | —                     | —                    | —                    | 9                          |
| L1:L2:L3(:L4)%            | 40:40:20                                   | 41:41:18              | 40:40:20             | 40:40:19             | 36:31:21:12                |
| Initial station           | Uniform over all stations in each scenario |                       |                      |                      |                            |
| Rate L1 (\$/h)            | 100                                        | 100                   | 100                  | 100                  | 100                        |
| Rate L2 (\$/h)            | 300                                        | 300                   | 300                  | 300                  | 300                        |
| Rate L3 (\$/h)            | 500                                        | 600                   | 500                  | 500                  | 500                        |
| Rate L4 (\$/h)            |                                            |                       |                      |                      |                            |
| Order demand distribution |                                            |                       |                      |                      |                            |
| Order level L1            | 40%                                        | 70%                   | 40%                  | 35%                  | 30%                        |
| Order level L2            | 40%                                        | 20%                   | 40%                  | 35%                  | 25%                        |
| Order level L3            | 20%                                        | 10%                   | 20%                  | 30%                  | 25%                        |

| Parameter        | S1                                                               | S2              | S3                                      | S4              | S5              |
|------------------|------------------------------------------------------------------|-----------------|-----------------------------------------|-----------------|-----------------|
| Order level L4   | —                                                                | —               | —                                       | —               | 20%             |
| Pick-up station  | Uniform, all 14                                                  | Uniform, all 16 | Stns 1-4: 48%; stns 5-18: 52% uniform   | Uniform, all 20 | Uniform, all 20 |
| Return station   | Uniform, all 14                                                  | Uniform, all 16 | Stns 15-18: 48%; stns 1-14: 52% uniform | Uniform, all 20 | Uniform, all 20 |
| Return time      | Must end within planning horizon; resampled if exceeded          |                 |                                         |                 |                 |
| Moving time step | Multiple of 30 min; $T_{ij} = T_{ji}$ ; $T_{ii} = 0$ for all $i$ |                 |                                         |                 |                 |
| Horizon start    | 2023/01/01 00:00 for all scenarios                               |                 |                                         |                 |                 |

Blue = parameter changed from S1 or notably distinct. Orange ■ = parameter inconsistent with scenario purpose (S4 order/car ratio lower than S1; S3 and S4 budget may be too generous). S2 uses rate 600 for level-3 (vs 500 elsewhere). Seeds follow pattern  $4900 + \text{scenario\_index} \times 100 + \text{instance\_index}$ .